A Shape and Size Free-CNN for Urban Functional Zone Mapping With High-Resolution Satellite Images and POI Data

Zhou Guo, Jiangtian Wen, and Rui Xu

***Abstract*— Urban functional zone (UFZ) refers to the spatial aggregation of similar human activities in urban areas, and its category information has significant implications for city planning and layout. Existing studies have incorporated high spatial resolution (HSR) remote sensing images with social sensing data to obtain UFZ patches for classification and identification purposes. While deep learning techniques have proven effective in remote sensing image classification, two challenges arise when applying them to UFZ classification: irregular shapes and inconsistent sizes, making it difficult to input UFZ patches into deep learning models directly. To address these challenges, this study proposes an end-to-end model, known as the shape and size free convolutional neural network (SSF-CNN), to automatically classify UFZ patches of varying sizes and irregular shapes. First, the SSF-CNN adopted a novel network, named hierarchical attentional residual network (Res-HANet), which embeds a hierarchical group convolution (HGC) module and attention mechanisms to learn multiscale features from fused image blocks of four different sizes. Then, a mask layer is followed to filter the deep features and preserve the original information of irregular UFZs. The proposed method was applied to classifying UFZs in Zhuhai and Guangzhou, Guangdong, China. The evaluation results showed that SSF-CNN achieved an overall accuracy (OA) of 87.85% for the Zhuhai dataset and 90.49% for the Guangzhou dataset, significantly better than existing methods. In addition, ablation experiments confirm the effectiveness of components in the SSF-CNN. Overall, the results suggest that the proposed method has great potential for large-scale UFZ mapping.**

***Index Terms*— Deep learning, high spatial resolution (HSR) remote sensing images, irregular unit, multiscale, urban functional zone (UFZ).**

I. Introduction

IN URBAN areas, land use with similar characteristics often necessitates the same spatial allocation and location, giving rise to functional zones such as residential, commercial, and industrial zones [1]. Classifying urban functional zones (UFZs) enables a rational urban layout, which is vital for promoting productivity, enhancing the quality of life of residents, and protecting the urban environment. Traditional qualitative methods of UFZ classification, including statistical surveys and expert knowledge [2], are limited by statistical data and subjectivity, making it challenging to establish the precision of results while being time-consuming and labor-intensive.

High spatial resolution (HSR) remote sensing images have the advantages of providing detailed information about ground objects, covering a wide range of spaces, and updating timely. Therefore, they are widely used for geographic scene recognition [3]. For instance, methods, such as scale-invariant feature transform (SIFT) and histogram of oriented gradient (HOG), have been utilized to extract shallow visual features of HSR images to identify land use and land cover (LULC) [4], [5]. However, these shallow visual features can only be used for identifying simple scenes with unique visual symbols and are challenging to classify heterogeneous scenes such as UFZs with multiple objects and complex visual characteristics. Theme models (TM) can solve complex heterogeneous scenario classification to a certain extent. For example, the bag of visual words (BOVW) model considers UFZs as a series of "visual words," where each word represents a typical visual pattern [6] and encodes the characteristics of UFZs through these visual words. The latent Dirichlet allocation (LDA) and probabilistic latent semantic analysis (pLSA) models further employ object features and categories to generate latent semantics, enabling the description of image scenes as different UFZs [7], [8].

While HSR images are valuable resources for identifying the physical attributes of cities, they do not offer insights into functional interaction patterns and socioeconomic environments, which also constitute crucial features of UFZ. As a remedy, scholars have suggested the incorporation of social sensing data, encompassing mobile phone signaling data, taxi trajectory data, points of interest (POIs), and so on. For instance, Zhong et al. [9] inferred building-level UFZ from traffic data obtained from a smart card system based on a probabilistic Bayesian model. Jiang et al. [10] proposed a method for identifying UFZ based on mobile phone signaling data, leveraging the correspondence between UFZ categories and mobile phone users' behavioral characteristics. Among all the social sensing data, POI has a high degree of matching with urban land use types [11], which explains its popularity in UFZ classification [12]. The current approaches employed in handling POI encompass a range of techniques. These primarily comprise density analysis, for instance, kernel density [13], point density [14], and distance heat map [15].

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Density analysis quantifies the spatial dispersion of POIs with varied functional attributes and serves as a basis for identifying UFZs. Another technique used is cluster analysis, such as fuzzy c-means (FCM) clustering [16] and logistic regression model [17]. Cluster analysis places more emphasis on the spatial correlation between POIs, thus enabling the identification and classification of UFZ based on the obtained clustering outcomes.

Recent years have witnessed tremendous advancements in deep learning techniques. Compared to traditional methods that rely on manual features or models with shallow architectures, deep learning models exhibit superior feature extraction capabilities, making them ideal for handling complex tasks such as UFZ classification. Convolutional neural networks (CNNs) have been successfully applied in remote sensing image processing [18]. Many studies have experimented with CNNs to extract high-level visual features from HSR images for UFZ classification. For instance, Bao et al. [19] proposed a deeper feature CNN (DFCNN) model, which can extract more and deeper features and has been successfully applied in the semantic recognition of UFZs by integrating HSR images and POI data. Residual network (ResNet), which uses residual blocks to implement a network with deeper layers for feature extraction [20], has also been exploited as a feature extraction module in many UFZ classification networks [15], [21], [22]. In contrast to CNNs specializing in processing images, recurrent neural networks (RNNs) can capture the contextual features of input data. Yao et al. [23] utilized the traffic analysis zones (TAZs) as the basic UFZ classification units. They analyzed the spatial relationship between TAZs and POIs within the Word2Vec model to detect the distribution of UFZs accordingly. On the other hand, some social sensing data also have temporal characteristics, such as pedestrian travel data. RNNs, such as long short-term memory (LSTM) networks, can extract the temporal features of such data as a basis for UFZ discrimination [21]. Furthermore, graph-based approaches are emerging as a popular method for exploring the spatial relationships of ground objects. Yang et al. [24] constructed a graph of buildings inside UFZs and proposed a graph CNN with discriminable pooling (DGCNN) to extract morphological features within UFZs. Wang et al. [25] considered human mobility among UFZs, converted UFZs into nodes in mobility graphs, and employed multisource semantic features inside the UFZs as embeddings of each node. The UFZs were then identified using the graph sample and aggregate (SAGE) model.

Deep learning methods have shown exemplary performance in UFZ classification. However, two inherent contradictions exist when applying deep learning to UFZ classification. First, deep learning models only support regular square image inputs, whereas UFZs often have arbitrary shapes. Second, deep learning models usually require the input images to be the same size when performing batch processing. However, the sizes of UFZs are often not uniform. Previous research has typically addressed these issues using a "sampling" approach [6], [25], [26]—setting a sampling window of a specific size and cropping irregular UFZs to obtain small, uniformly sized square patches. These patches are then fed into the models for training and prediction, with predicted categories mapped onto the original UFZs. However, this preprocessing method fragments the UFZs and makes it difficult for the model to extract features from the complete picture of each UFZ, while determining the optimal size for the sampling window is also challenging.

This study proposes an end-to-end deep learning model called shape and size free-CNN (SSF-CNN), which leverages fused image data from HSR images and POIs. The input of SSF-CNN is a square image block cropped out of the square bounding box (SBB) of each UFZ. These image blocks are then resampled to four sizes at multiple resolutions and input into a novel deep network, which can learn multiscale features from multisize input images, thereby solving the problem of UFZ size inconsistency. A mask layer is followed to filter out extraneous features outside the UFZ, retaining only the features within the UFZ, thereby solving the irregular shape problem. The resulting features are then utilized for UFZ classification. The main contributions of this study are given as follows.

1) We propose a multiscale feature learning method, named hierarchical attentional residual network (Res-HANet), that maximizes the retention of scale information of UFZs of varying sizes. The proposed method processes input images at multiple sizes while following batch processing rules.
2) We design a novel network structure, i.e., the mask layer, which filters out extraneous features outside the irregular UFZ, enabling the processing of irregular UFZs in a deep network.
3) The proposed SSF-CNN is an end-to-end deep learning model that accepts the overall input of UFZ images without additional data preprocessing operations. Compared with other UFZ classification models, it is easier to use, more efficient to execute, and can achieve better classification results.

The remainder of this article is organized as follows. Section II provides a review of the related works. Section III describes the study area, datasets, and definitions of UFZ. Next, the proposed SSF-CNN is explained in Section IV. Section V presents the experimental settings, UFZ mapping results, and comparisons, followed by a discussion in Section VI. Finally, we conclude this article in Section VII.

II. Related Works

UFZ classification methods with HSR imagery based on deep learning can be categorized into three types: grid-based methods, sampling-based methods, and graph-based methods.

A. Grid-Based Methods

Grid-based methods are primarily based on dividing HSR images into grids or squared patches that are set as the basic UFZ units and then utilizing deep learning models to classify each UFZ. For example, Lu et al. [15] employed a uniform division method to divide their study area into grids of 15×15 m^2, which were set as the basic UFZ units.

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Two coupled CNNs were used to learn visual features from HSR imagery and social features from POIs. A feature attention-based fusion strategy was then designed to fuse the complementary features with adaptive learned weights in an end-to-end manner. However, researchers may argue that it is not suitable to represent UFZs of varying sizes with uniform grids. In order to alleviate the shortcoming, Jing et al. [27] adopted a hierarchical grid division strategy based on the information entropy of POIs. They initially partitioned the entire study area into larger grids and calculated the POI information entropy within each grid, which reflected the extent of functional type amalgamation within the grid. Whenever the POI information entropy exceeded a certain threshold, indicating high levels of mixture, the grid was further subdivided into smaller girds. The process iterated until all information entropy fell below the threshold. Consequently, the derived grids were severed as final UFZ units. Meanwhile, Cao et al. [21] developed a deep learning strategy to fuse social sensing features by an LSTM network and remote sensing image features by a residual neural network. They adopted image patches from a public benchmark dataset to train the deep network and classify UFZs. The greatest challenge of grid-based methods is the incapability of fully representing UFZs by regular grids due to the inherent complexity of urban environments.

B. Sampling-Based Methods

Sampling-based methods usually adopt blocks as basic UFZ units. However, since common deep learning methods do not accept irregular UFZs as inputs, different sampling strategies were used. Zhou et al. [26] proposed a super object-CNN (SO-CNN)-based UFZ classification method, in which HSR imagery was first divided into SOs corresponding to UFZs. A random point generation algorithm was used to generate points within UFZs, and a trained CNN model was employed to classify image patches, which used those points as image centers and assigned a functional label to each point. A majority voting strategy was involved to determine the final functional attribute of each UFZ. Aside from random point generation, other studies have generated points according to certain rules. For example, Huang et al. [28] proposed a deep fusion network (DFN), which also employed the sampling strategy for UFZ classification. The basic units of UFZs used in DFN were blocks divided by the OpenStreetMap (OSM) road network. Different from the random point sampling strategy in SO-CNN, a sliding window sampling method was used in DFN, in which a 130×130 m^2 window was utilized to generate points. The window started at the top left of a UFZ and slid from left to right and from top to bottom. Each time, the window slides 65 m, generating a small image patch. If the image patch has an overlap of more than 80% with the original UFZ, the image patch is retained and assigned by the DFN. Similarly, a majority voting method was adopted to determine the final label of a UFZ. Zhang et al. [29] proposed an equal-area division algorithm to generate sampling points. First, they segmented an HSR image into image objects. Second, an iterative sampling strategy based on equal-area division method was used to generate sampling points within each image object. Each time, an image object was divided into two subobjects of the same area horizontally or vertically according to the shape of the object. The centroid of the subobject was extracted. The above steps iterate until the number of centroid points reaches a preset number. Image patches were then generated at each sampling point and sent to CNN, and the same majority voting method was used to determine the label of an image object. Although the sampling-based methods have tactfully transformed the irregular UFZ classification problem into a regular patch classification case, the small sampled patches are incapable of representing the global information of a UFZ.

C. Graph-Based Methods

Graph-based methods employ a graph structure to represent UFZs and topological connections between UFZs. For example, Hu et al. [30] first constructed a graph of road segments, in which each node in the graph represented a road segment, and each graph edge represented the connection between two road segments. The semantic embedding of road segments was assigned to graph nodes as their attributes. Second, a graph CNN model was adopted to process the contextual and topological information between road segments and classify social functions along streets. Third, the function of a UFZ was determined by its surrounding road streets. Wang et al. [25] considered human mobility between UFZs by constructing a mobility graph. In the graph, each UFZ was represented by a node, and multisource semantic features inside UFZs were represented by embeddings of each node. Trajectory information between UFZs was represented by graph edges and reflected topological relations between UFZs. A graph SAGE method was adopted to identify UFZs. Other graph-based UFZ classification methods include [24], [31], and [32]. One significant advantage of graph-based methods is that they consider the topological relations between UFZs, which improves UFZ identification results in some cases. However, the graph-based methods require high standards for training samples of UFZs in terms of both quality and quantity, which is not easy to reach. Moreover, the multiscale information of UFZs is often neglected in graph-based methods.

III. Study Areas and Datasets

A. Study Areas

Two cities, Zhuhai and Guangzhou, located in Guangdong, China, were selected as study areas (see Fig. 1). Zhuhai (22°16′ N, 113°34′ E) is a prefecture-level city, which has a land area of 1736 km^2 and a reported resident population of 2.44 million in 2020. The city was established as a Special Economic Zone in 1980, and the resulting economic growth has made it one of the country's leading cities in GDP per capita. Guangzhou (23°8′ N, 113°16′ E), the capital of Guangdong, is a globally recognized megacity. Its municipal area covers 7434.4 km^2. As of 2021, the resident population of Guangzhou was recorded to be 18.8 million, with an urbanization rate of 86.46%. Zhuhai and Guangzhou have achieved significant economic development

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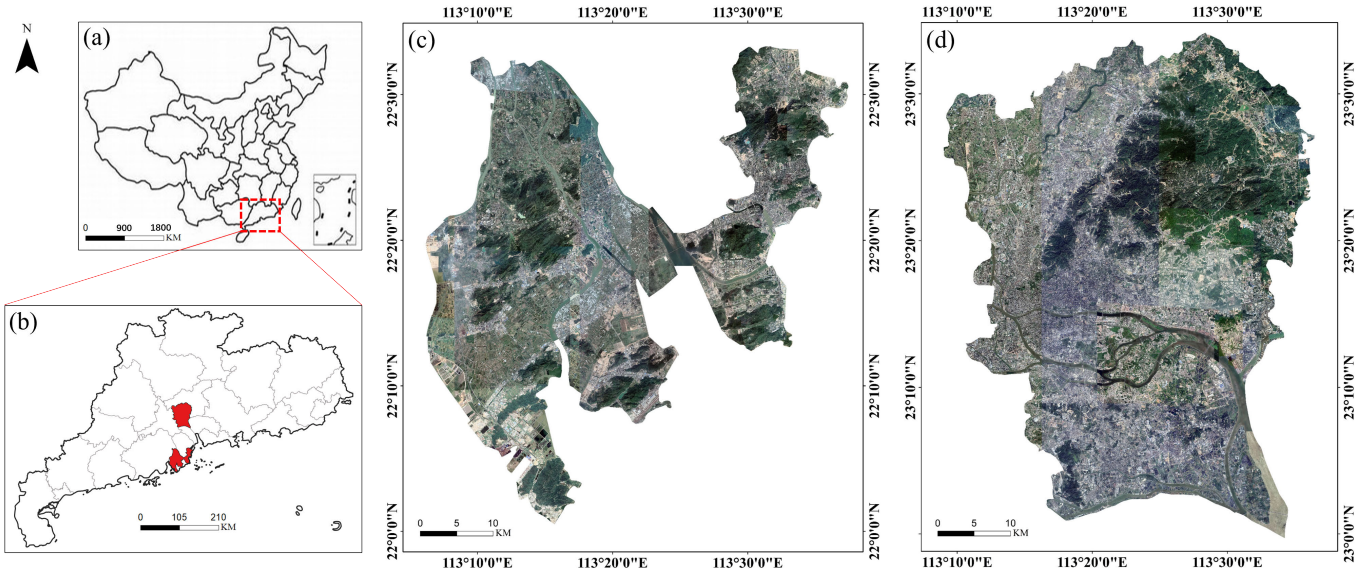


Fig. 1. Illustrations of study areas. (a) Map of China. (b) Map of Guangdong. (c) Zhuhai study area. (d) Guangzhou study area.

with balanced agricultural, industrial, and commercial growth. This has given rise to diverse urban land utilization and human activities. As a result, these cities present ideal conditions for the study of UFZ classification.

To facilitate subsequent research, only the land part in Zhuhai was kept [see Fig. 1(c)], and seven districts in the central part of Guangzhou were employed, namely, Haizhu, Yuexiu, Liwan, Tianhe, Panyu, Baiyun, and Huangpu [see Fig. 1(d)].

B. Definitions of UFZs

In this study, we used blocks as the basic unit of UFZs, which were divided by a network of roads. The road networks (see Fig. 2) were sourced from OSM and obtained in October 2022.

Prior to delineating the blocks, the original OSM network data [see Fig. 2(a)] underwent three preprocessing steps. First, in order to reduce oversegmentations of UFZs as well as avoid tiny zones that do not carry any meaningful socioeconomic functions, we removed the low-level roads such as internal roads and pedestrian paths. In this study, roads below the level of "tertiary" were regarded as low-level roads, which has referenced several existing studies [6], [25], [33]. Fig. 2(b) shows the remaining roads. Second, to further remove the redundant road details, such as lanes, ramps, and overpasses, roads were thickened by performing a dilation process. As a result, the redundant details were smoothed out and holes were filled [see Fig. 2(c)]. Third, to obtain road skeletons, a thinning operation was performed. Furthermore, topological errors, such as hanging lines, were corrected in the road network [see Fig. 2(d)]. After obtaining the processed road network, UFZs were defined as polygons enclosed by roads in the network.

As a result, a total number of 1543 and 4477 UFZ blocks were produced for Zhuhai and Guangzhou study areas, respectively. In consideration of the urbanization process of the two

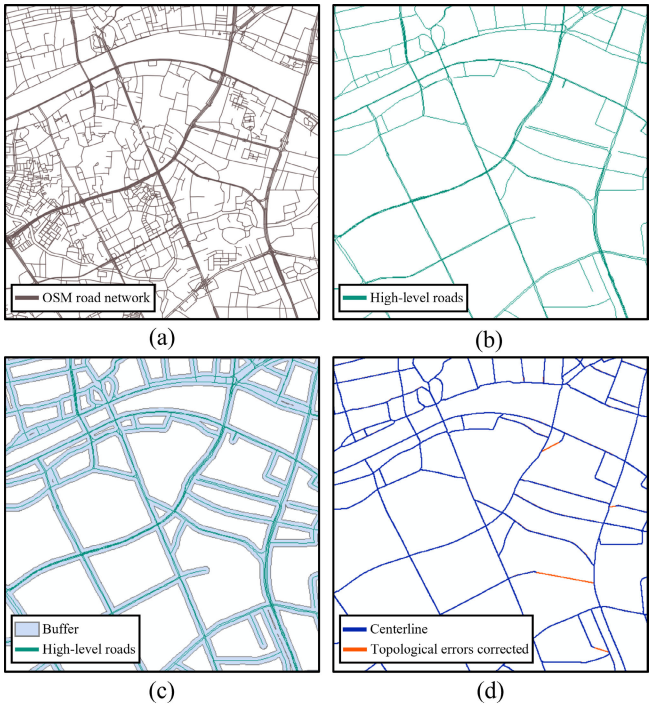


Fig. 2. Processes of generating UFZ blocks. (a) Original OSM road network, (b) remaining OSM roads after removals of low-level roads (below tertiary), (c) thickened roads by the dilation process, and (d) road skeletons after the thinning process and correcting topological errors.

cities as well as referring to government policies [34] and existing studies [6], [35], seven types of UFZs were defined: residential zones, public service zones, green space zones, industrial zones, commercial zones, educational zones, and agricultural zones. Their definitions are detailed in Table I.

In order to train and test our proposed model for UFZ mapping, categories of UFZs in the two study areas were manually labeled via fieldwork surveys and open-source geographic information such as street-view images.

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TABLE I
DEFINITIONS OF THE SEVEN TYPES OF UFZS IN THIS STUDY

Types	Definitions
Residential zones	Complex of buildings composed of various types of residences and related structures.
Public service zones	Areas for providing public services, such as government departments, hospitals, and stations.
Green space zones	Areas that have a positive effect on the ecology, landscape, and recreational life of the city and have a good green environment.
Industrial zones	Areas with a relatively concentrated layout of industrial enterprises and production activities.
Commercial zones	Areas with a relatively high concentration of commercial facilities, such as large shopping centers.
Educational zones	Areas with a relative concentration of schools and research institutions.
Agricultural zones	Areas for agricultural production.

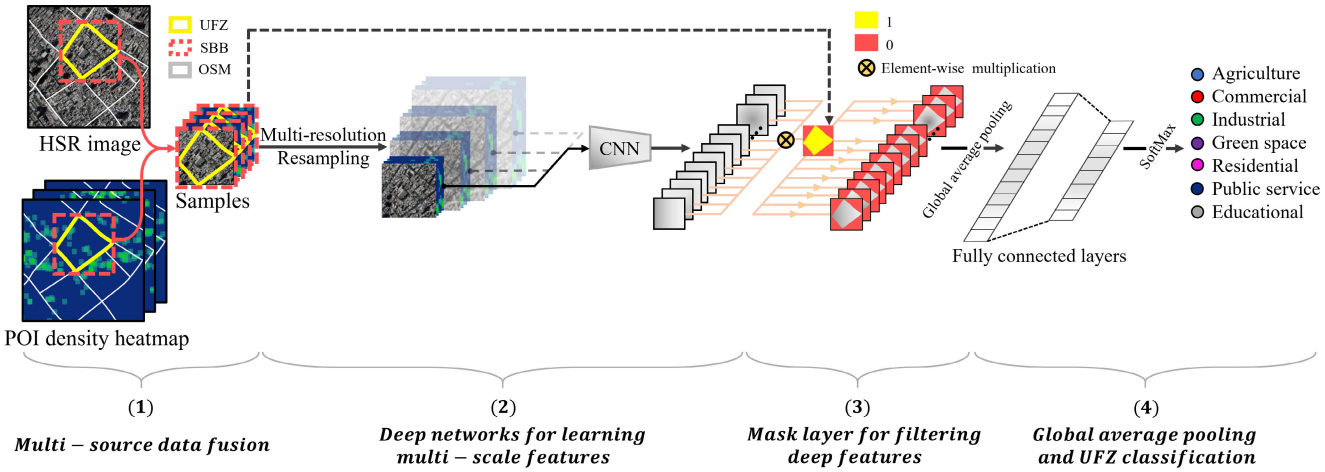


Fig. 3. Overall structure of SSF-CNN.

C. Datasets

Two types of data were employed for UFZ mapping: HSR images and POIs. The HSR images were downloaded from Google Earth in October 2022 with a resolution of 1 m and three bands: red, blue, and green. Furthermore, to supplement socioeconomic information, we extracted POIs from AutoNavi (https://lbs.amap.com/), a Chinese online map platform, in June 2022. UFZs encompassing commerce, education, and public service zones lack distinct visual characteristics that can be directly discerned from HSR images. Consequently, the integration of POI data becomes imperative to facilitate the identification process by supplementing socioeconomic information. Conversely, other UFZ categories exhibit unique visual attributes that obviate the necessity of utilizing POI data for identification purposes. To minimize data redundancy, only three categories of POIs were considered and kept, namely, commerce, education, and public service. A total of 15 395 and 55 891 points were obtained for the Zhuhai and Guangzhou datasets, respectively.

IV. METHODOLOGY

The structure of the proposed method is shown in Fig. 3, which mainly includes four parts: 1) multisource data fusion; 2) deep networks for learning multiscale features; 3) mask layer for filtering deep features; and 4) global average pooling and UFZ classification. The method is described in detail as follows.

A. Multisource Data Fusion

HSR images and POI data are complementary to each other and contribute to UFZ mapping in different ways. However, fusing the two data sources is difficult as they have different data structures. To solve this problem, we convert discrete POIs into a multichannel point density map, in which each channel represents the density distribution of a POI class. As we used three types of POIs, a three-channel heatmap was generated. Specifically, for each pixel i in the kth ($k = 1, 2, 3$) channel of the heatmap, its pixel value is defined by the point density [36] of the kth POI type within a certain neighborhood

$$\text{Density}_i^k = \frac{N_i^k}{\text{Area}_i} \tag{1}$$

where Area_i is the area of the neighborhood for pixel i and N_i^k is the number of POIs of the kth type within the neighborhood. In this study, a radius of 300 pixels is set for defining the neighborhood.

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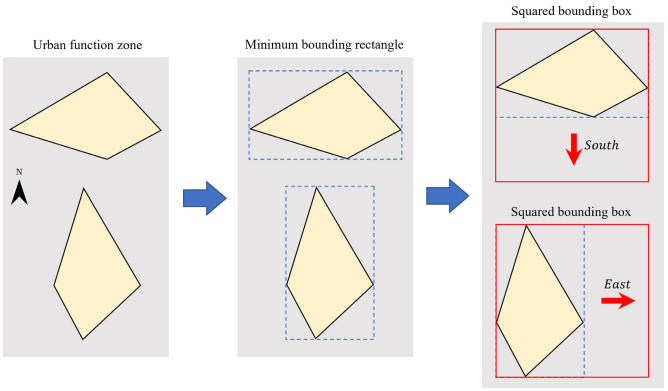


Fig. 4. Steps for generating an SBB.

The obtained three-channel density map was sampled to the same resolution as the HSR image and then stacked with the HSR image to form a fused map. The fused map can be expressed as $\mathbf{I} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W represent the number of channels, width, and height of the fused map, respectively. In this study, C is six and varies as the numbers of bands of HSR images and types of POIs change.

After data fusion, the fused map was divided by SBBs of UFZs. Generating an SBB for a UFZ is completed in two steps (as shown in Fig. 4). First, a minimum bounding rectangle (MBR) is generated for each UFZ. Second, the MBR is enlarged toward the south or east direction to form a square, i.e., the SBB.

After the fusion map is divided by the SBBs, the output is a set of squared image blocks and a set of masks representing the boundaries of UFZs, which are in a one-to-one correspondence (as shown in Fig. 5). It is worth noting that the sizes of squared image blocks vary as UFZs have different sizes. The squared image blocks and masks are then served as input to our proposed model in Sections IV-B and IV-C.

B. Deep Networks for Learning Multiscale Features

Most existing CNNs only accept input images that are squared and in a uniform size, which is obviously against the characteristics of UFZs. In order to learn features of UFZs that are irregular and with varying sizes, a novel CNN-based deep network is proposed, called Res-HANet.

1) Multiresolution Resampling of Squared Image Blocks: CNN-based deep neural network often requires fixed-size images as inputs [37]. Therefore, it is necessary to resize the squared image blocks in Section IV-A. However, these image blocks vary significantly in size, and it will cause a serious scale effect if they are resampled to a uniform size [38]. To reduce the scale effect, all squared image blocks are resampled to four different sizes, i.e., 192×192, 256×256, 512×512, and 800×800. The four sizes are chosen as they are distributed as evenly as possible within the limits of our hardware, and such a design can guarantee that the scale information of a UFZ with an arbitrary size can be captured by the four sizes. After resampling, the number of image blocks expands four times, and they are all fed into our network for feature learning.

2) Overall Architecture of the Feature Learning Network: The structure of the feature learning network is shown in Fig. 6(b). The architecture consists of five stages, with the first stage comprising a 7×7 convolutional layer and a 3×3 max-pooling layer, and the subsequent four stages comprising several residual blocks, called hierarchical attentional residual (Res-HA) module. The whole network Res-HANet is very similar to the configuration of ResNet [20]. Comparatively, ResNet-50 contains three, four, six, and three standard residual blocks [see Fig. 7(b)], while our network contains three, four, six, and three Res-HA modules [see Fig. 7(a)]. As shown in Fig. 7(a), a Res-HA module has a ResNet structure. It contains a 1×1 convolution, a hierarchical group convolution (HGC) for multiscale feature learning, and a small convolutional block attention module (sCBAM) for attention mechanism, followed by another 1×1 convolution. Assume that the input to a Res-HA module is x, and the output of Res-HA module can be represented as

$$y = x + \text{conv}^{1\times1}\left(\text{sCBAM}\left(\text{HGC}\left(\text{conv}^{1\times1}(x)\right)\right)\right) \tag{2}$$

where $\text{conv}^{1\times1}(\cdot)$ is an 1×1 convolutional operation, $\text{HGC}(\cdot)$ is the multiscale feature learning operation, and $\text{sCBAM}(\cdot)$ is the attention mechanism operation. The latter two operations will be introduced in Sections IV-B3 and IV-B4.

The Res-HA module itself does not change the size of input images. However, the size of input images will be reduced to half by convolutional layers with a stride of 2. Therefore, the size of the output images will be decreased to 1/32 of the original inputs. As the dimensions of input images of Res-HANet are $6 \times 192 \times 192$, $6 \times 256 \times 256$, $6 \times 512 \times 512$, and $6 \times 800 \times 800$, the dimensions of outputs by the feature learning network are $2048 \times 6 \times 6$, $2048 \times 8 \times 8$, $2048 \times 16 \times 16$, and $2048 \times 25 \times 25$, respectively.

3) Multiscale Feature Learning With HGC: In order to learn multiscale features from input images, we propose an HGC module that references the architecture of Res2Net [39]. As can be seen in Fig. 7(a), HGC first divides the input feature map x equally into four subset images in terms of channels, i.e., x_1–x_4. Such a division guarantees that no information is lost during the operation, as the original feature map can be easily restored. Then, the above four images are processed hierarchically. For x_1, no operation is performed, and the output y_1 is the same as its input x_1. For x_2, a 3×3 convolution $\text{conv}^{3\times3}(\cdot)$ is performed, and its output y_2 is $\text{conv}^{3\times3}(x_2)$. For x_3 and x_4, the image is first added with its former output feature, and a 3×3 convolution is then performed on the added feature map. Therefore, the outputs of the four images can be represented as

$$\mathbf{y}_i = \begin{cases} \mathbf{x}_i, & i = 1; \\ \text{conv}^{3\times3}(\mathbf{x}_i), & i = 2; \\ \text{conv}^{3\times3}(\mathbf{x}_i + \mathbf{y}_{i-1}), & 2 < i \leq 4. \end{cases} \tag{3}$$

According to (3), the larger the subscript number is, the more convolutions are performed, and the larger perceptual fields can be sensed [40], resulting in higher level

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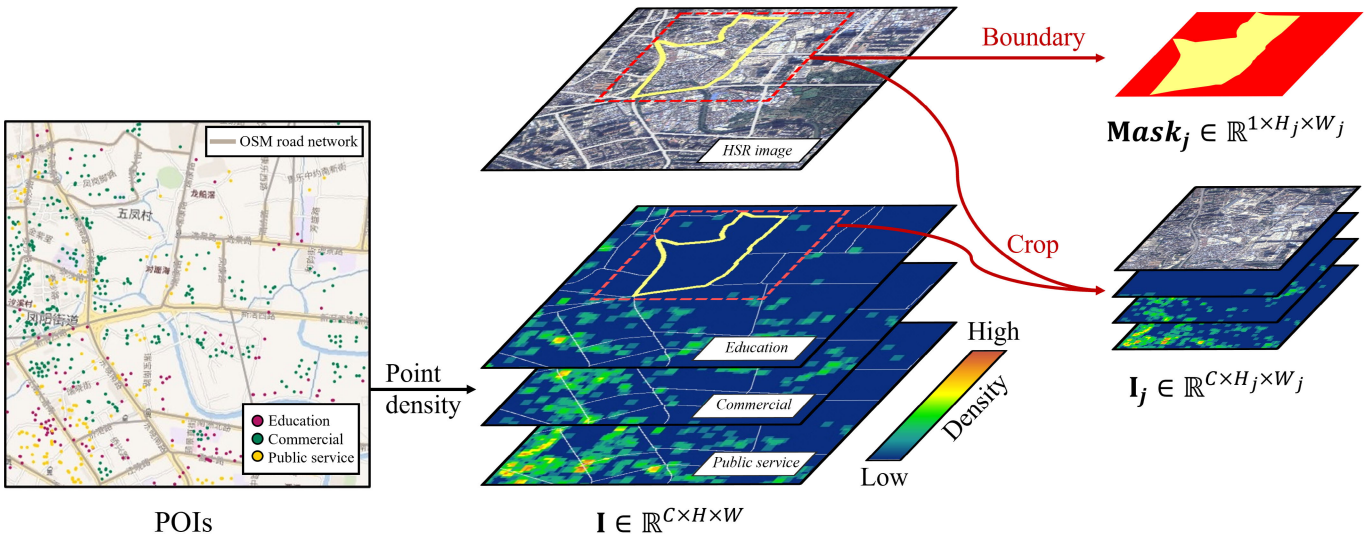


Fig. 5. Process of generating the samples.

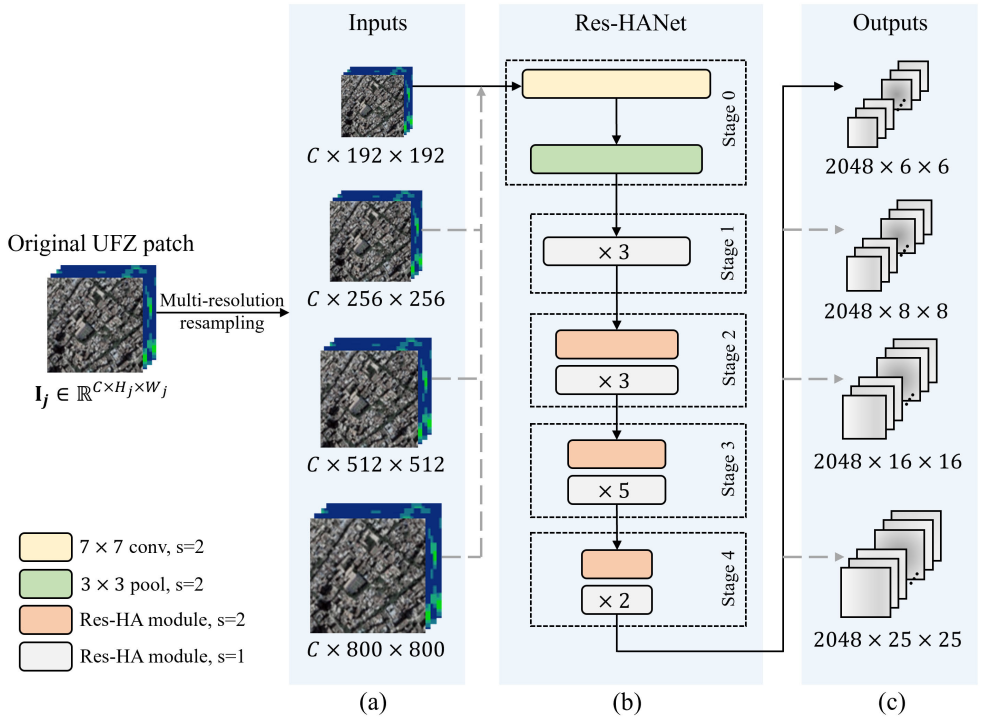


Fig. 6. Deep network Res-HANet with different sizes of inputs and outputs. (a) Inputs of Res-HANet. (b) Overall structure of Res-HANet. (c) Outputs of Res-HANet.

feature maps. The four output feature maps, i.e., y_1–y_4, are concatenated to fuse all their information, resulting in multi-scale feature maps with different levels. The results are then sent to the sCBAM module introduced in Section IV-B4.

4) Attention Mechanism in sCBAM: The multiscale features learned in Section IV-B3 through the HGC module can be conceptualized as having two dimensions: channel and spatial. In the channel dimension, different channels of the fused image contribute differently to UFZ classification, and a channel attention mechanism can assign weights to channels and refine features in channels. In the spatial dimension, a UFZ is usually composed of several land uses in different regions, and a spatial attention mechanism can refine features adaptively in space. In order to refine the spatial and channel features and enhance the performance of the proposed Res-HANet, we present an enhanced version of CBAM [41], called sCBAM. Specifically, the fully connected layer of the original CBAM module has been replaced with a 1-D convolutional layer, resulting in a reduction in parameter size [42]. The sCBAM is placed after the HGC module and consists of two submodules: the channel and spatial attention modules (see Fig. 8).

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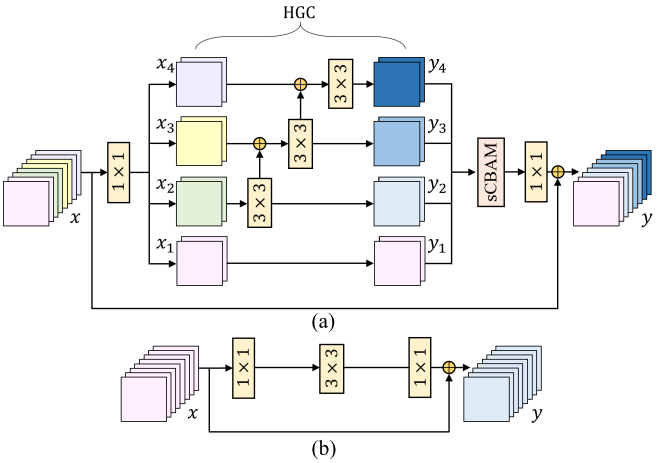


Fig. 7. Comparisons of the residual block of ResNet with that of Res-HANet. (a) Res-HANet residual block. (b) ResNet residual block.

In the channel attention module, the feature map is first subject to average-pooling operation in each channel, yielding a 1-D feature vector. Each value in this vector characterizes the global distribution of the corresponding channel in the feature map. The next step involves cross-channel interaction via 1-D convolution, capturing channel correlations to obtain a new 1-D feature vector. The kernel size of the 1-D convolution is adaptive to the number of channels and calculated as

$$k = \psi(c) = \left| \frac{\log_2 c}{2} + \frac{1}{2} \right|_{\text{odd}} \tag{4}$$

where k is the kernel size, c is the number of channels for the input feature map, and $|t|_{\text{odd}}$ indicates the nearest odd number of t. The new 1-D feature vector is then sent to a sigmoid activation function, which outputs a channel attention map encompassing importance weights for each channel. The channel attention map is multiplied by the input feature map for refining features in the channel dimension.

After the channel attention module, the feature map is sent to the spatial attention module. First, average- and max-pooling operations are separately applied to the feature map along the channel dimension, yielding two feature maps, which are then concatenated together. Second, a 7×7 convolution is performed on the concatenated feature map to mine the spatial correlations, and a sigmoid activation function is followed. It outputs a spatial attention map, which is further multiplied by the input feature map of the spatial attention module. By doing so, the input feature map is further refined in the spatial dimension.

C. Mask Layer for Filtering Deep Features

The above deep network extracts deep features from images of SBBs. Although an SBB encompasses a UFZ, its extent is larger than its corresponding UFZ. In order to obtain the deep features of UFZs, the feature maps of SBBs are further filtered by a mask layer (see Fig. 9), followed by the deep networks. The mask layer consists of a matrix of the same size as the output feature map of Section IV-B2. The matrix is used to store the shape information of the irregular UFZ. If a pixel in the feature map falls outside the mask, the corresponding element value in the matrix is set as 0; otherwise, the corresponding element value is set as 1. The feature maps of SBBs are element-wise multiplied with the matrix in the mask layer to filter features of UFZs, and irrelevant features are thus removed.

It is worth mentioning that the mask layer should be put after the deep networks instead of before the deep networks. This is because such a design allows the SSF-CNN to exploit image information around masks so that contextual features of UFZs can be learned by our proposed deep networks. Such an inference will be further verified by experiments demonstrated in Section VI-B4.

D. Global Average Pooling and UFZ Classification

The multiscale feature learning mechanism in Section IV-B outputs the feature maps of different sizes ($2048 \times 6 \times 6$, $2048 \times 8 \times 8$, $2048 \times 16 \times 16$, and $2048 \times 25 \times 25$). In order to uniformize their sizes, a global average-pooling operation is performed. All feature maps are transformed into 1-D feature vectors ($2048 \times 1 \times 1$) and then sent to a classification module consisting of a fully connected layer and a softmax layer. It outputs probabilities of each UFZ belonging to each category (see Fig. 10). The entire network optimization process is accomplished by minimizing the cross-entropy loss function

$$\text{Loss} = -\sum_{i} \sum_{k=1}^{C} y_{i,k} \log\left(p_{i,k}\right)$$

where $y_{i,k}$ and $p_{i,k}$ are the ground truth after one-hot coding and predicted probability value of the kth UFZ category for the ith sample, respectively, and C is the total number of UFZ categories.

During the network training, all training samples were resampled to four sizes illustrated in Section IV-B1, and a sequential approach was adopted in the training process. At each epoch, training samples of the same size were fed into the network and weights were updated. At the next epoch, training samples of another size were served as inputs. This process iterates until a certain number of epochs are reached. During the testing phase, there is no need to resample images to different sizes. Image blocks of original sizes were directly fed into the network for making predictions.

V. Experiments and Results

A. Experimental Settings

1) Implementation Details: In Zhuhai and Guangzhou datasets, 80% of samples are randomly selected for training, with the remaining 20% reserved for testing purposes. During training, samples with varying sizes are input into the model alternately and in random order. The number of epochs for each dataset is 240. Both datasets exhibit an imbalance of samples across categories, so data enhancement techniques

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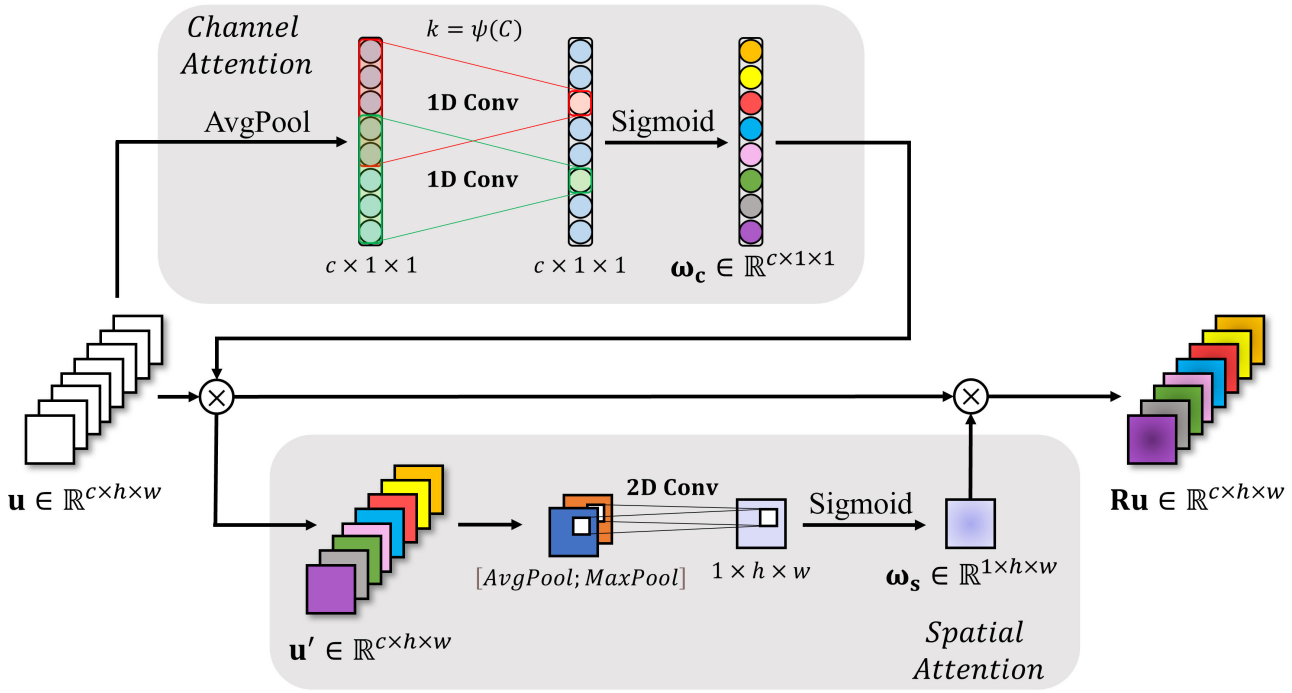


Fig. 8. Structure of sCBAM. It consists of two submodules: the channel attention module and the spatial attention module.

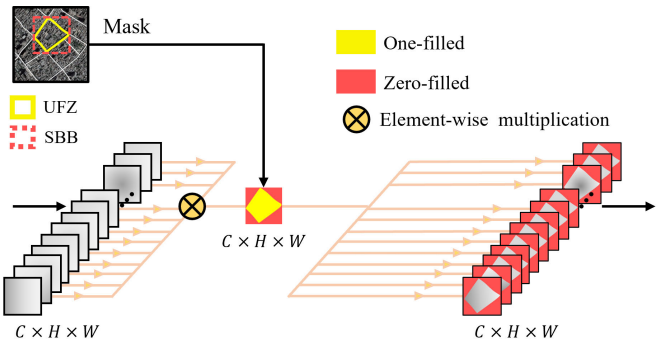


Fig. 9. Working principle of mask layer.

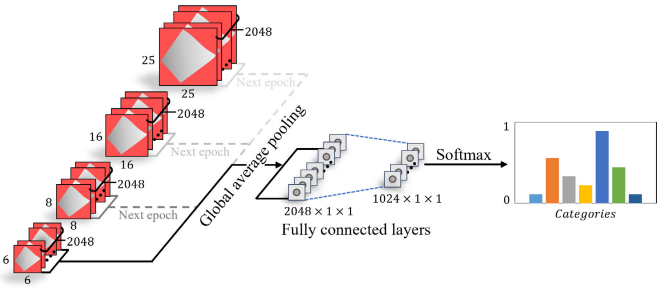


Fig. 10. Global average pooling and UFZ classification.

are applied to categories with limited training samples. Methods utilized include: 1) random rotation; 2) mirror flip; and 3) adding a small quantity of salt and pepper noise. The number of samples before and after data enhancement is shown in Table II.

The SSF-CNN proposed here is trained from scratch in an end-to-end format devoid of pretraining weights. The Adam optimizer and the cross-entropy loss function are employed during the training process, with a learning rate of 0.0001 being set. All experiments were conducted on Ubuntu 20.04.4, with the models being constructed using the Pytorch framework. The hardware utilized in this study includes two GPUs of Nvidia RTX 3090 and a CPU of AMD Ryzen ThreadRipper 3990X. The elapsed time for training the Zhuhai dataset and Guangzhou dataset is about 5.3 and 7.0 h, respectively.

2) Evaluation Metrics: The confusion matrix is an established convention utilized in presenting accuracy assessments. The rows correspond to the genuine categories to which the data pertains, and the entries in each column signify the number of predictions allocated to that category. The two primary measures of accuracy evaluation derived from the confusion matrix are overall accuracy (OA) and Kappa coefficient (Kappa), widely employed in practice. Furthermore, the assessment of classification accuracy for every UFZ category is conducted through the utilization of the $F1$ scores. They are calculated as follows:

$$\begin{cases} \mathrm{OA} = \dfrac{\sum_{i=1}^{r} x_{ii}}{N} \\ \mathrm{Kappa} = \dfrac{N \sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})}{N^2 - \sum_{i=1}^{r} (x_{i+} \cdot x_{+i})} \\ F1 = 2 \dfrac{x_{i1} \cdot x_{1i}}{x_{i1} \cdot x_{+i} + x_{1i} \cdot x_{i+}} \end{cases} \tag{5}$$

where x_{ii} is the value of row i and column i in the confusion matrix, which represents the number of samples in

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TABLE II
NUMBER OF SAMPLES BEFORE AND AFTER DATA ENHANCEMENT IN ZHUHAI AND GUANGZHOU

	Study area	Agriculture	Commercial	Industrial	Green space	Residential	Public service	Educational	Total
Before	Zhuhai	115	173	439	219	435	53	109	1543
	Guangzhou	515	363	968	634	1213	211	573	4477
After	Zhuhai	345	346	439	438	435	318	436	2757
	Guangzhou	515	726	768	634	713	633	573	4562

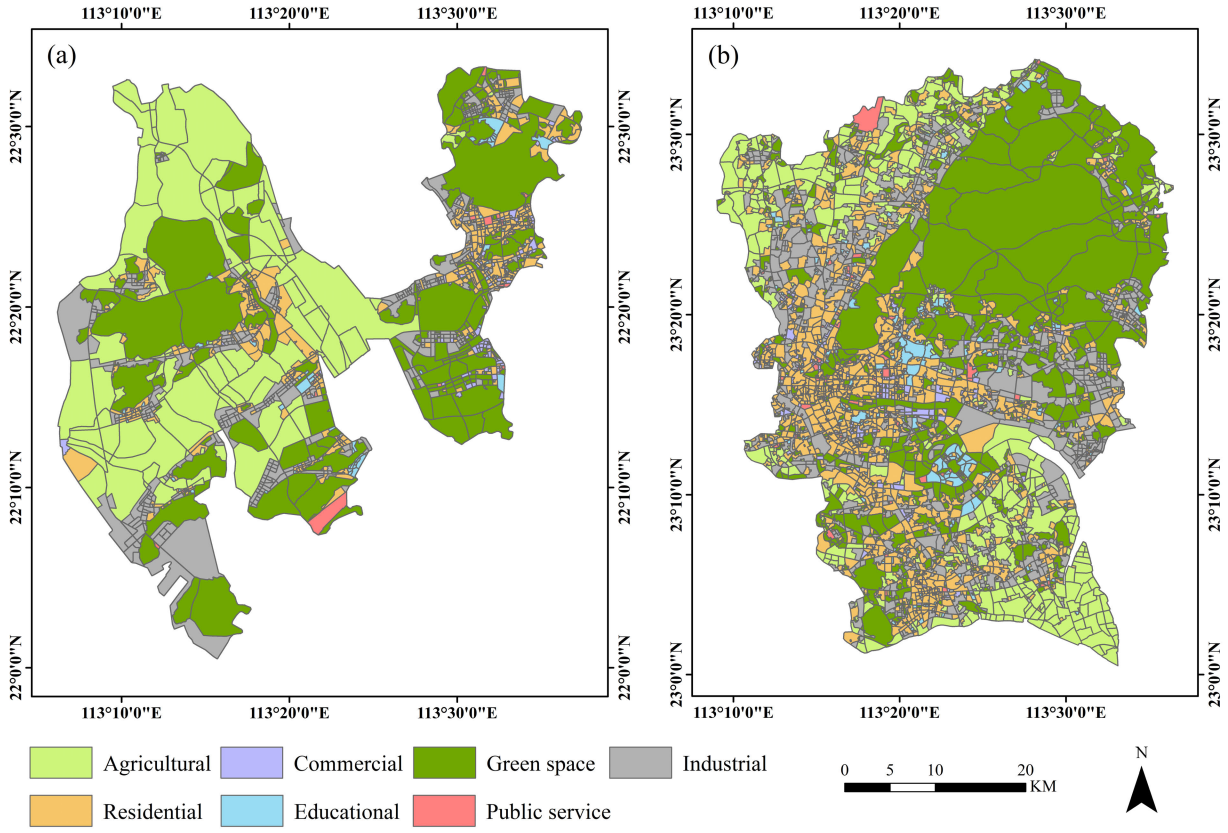


Fig. 11. UFZ classification maps based on SSF-CNN. UFZ map of (a) Zhuhai and (b) Guangzhou.

category i that were correctly classified; r is the number of rows (or columns) of the confusion matrix, which represents the total number of categories; N is the total number of samples; and x_{i+} and x_{+i} are the summation of elements in row i and column i in the confusion matrix, respectively.

B. UFZ Mapping Results

The UFZ classification results for Zhuhai and Guangzhou are, respectively, shown in Fig. 11(a) and (b), and Table III provides a quantitative description of the classification results for both datasets.

1) Classification Results of the Zhuhai Dataset: As can be seen from Fig. 11(a), the results of UFZ mapping in Zhuhai are satisfactory overall. Moreover, the distribution of UFZs exhibits a particular regional distribution pattern, and the same UFZs have a tendency to cluster. Specifically, agricultural zones are primarily located in the western part of Zhuhai, interspersed with green space, industrial, and residential zones. In the northeast, apart from a few green space zones composed of mountains, commercial, residential, and public service zones, densely populate the region, indicating that the northeast is an urban center of Zhuhai.

Aside from the qualitative visualizations, quantitative evaluations in Table III also revealed the excellent performance of our proposed method. SSF-CNN attains an OA of 87.85% and a Kappa of 84.68% on the Zhuhai dataset. Moreover, concerning classification accuracy for each UFZ category, educational zones reached the highest $F1$ of 94.44%, while 88.89%, 82.76%, 89.53%, 83.50%, 88.59%, and 90.91% reached for agricultural, commercial, industrial, green space, residential, and public service zones, respectively. The high individual $F1$ scores indicate that SSF-CNN exhibits great fitting ability for each UFZ category.

2) Classification Results of the Guangzhou Dataset: Fig. 11(b) shows the distribution of UFZs in the Guangzhou

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TABLE III
QUANTITATIVE CLASSIFICATION RESULTS OF ZHUHAI AND GUANGZHOU DATASETS

Datasets	F1 (%)							OA (%)	Kappa (%)
	Agricultural	Commercial	Industrial	Green space	Residential	Public service	Educational		
Zhuhai	88.89	82.76	89.53	83.50	88.59	90.91	94.44	87.85	84.68
Guangzhou	86.91	90.79	91.15	85.71	94.12	88.10	94.12	90.49	88.60

TABLE IV
QUANTITATIVE COMPARISONS IN THE ZHUHAI DATASET

Method	F1 (%)							OA (%)	Kappa (%)
	Agricultural	Commercial	Industrial	Green space	Residential	Public service	Educational		
SO-CNN	83.34	69.97	70.66	57.89	70.03	71.00	82.05	72.25	66.09
DFN	85.71	81.25	75.41	55.17	76.77	85.71	84.21	77.63	72.69
MSSN	79.17	76.36	84.81	80.52	88.04	88.89	**97.67**	85.20	80.99
SPP-net	78.26	60.71	68.75	69.88	79.00	80.00	79.07	73.36	66.00
SSF-CNN	**88.89**	**82.76**	**89.53**	**83.50**	**88.59**	**90.91**	94.44	**87.85**	**84.68**

study area, which seems denser and more fragmented than in Zhuhai. This may be attributed to human activities in Guangzhou being more frequent and intense than in Zhuhai. The green space zones are primarily located in hilly regions in the northeast, while the agricultural zones mainly occupy the northwest and southeast fringes of the city.

Quantitatively, SSF-CNN achieved better classification results on the Guangzhou dataset than on the Zhuhai dataset. Specifically, the OA and Kappa of the Guangzhou dataset were 90.49% and 88.60%, respectively, 2.64% and 3.92%, higher than those of the Zhuhai dataset. Regarding individual categories, $F1$ of agricultural, public service, and educational zones in the Guangzhou dataset was 86.91%, 88.10%, and 94.12%, respectively, slightly lower than those in Zhuhai, while $F1$ of commercial, industrial, green space, and residential zones was 90.79%, 91.15%, 85.71%, and 94.12%, respectively, which were higher than those of Zhuhai. Educational and residential zones ranked the highest in the $F1$ score, i.e., 94.12%, while green space ranked the lowest in the $F1$ score, i.e., 85.71%.

C. Comparisons With Other Existing Methods

To assess the efficacy of the proposed SSF-CNN, it was compared with four existing deep learning-based methods with excellent UFZ classification performance, i.e., SO-CNN [26], DFN [28], multiscale semantic network (MSSN) [22], and spatial pyramid pooling network (SPP-net) [38], [43]. Specifically, SO-CNN and DFN are two sampling-based methods, which have been introduced in Section II, while MSSN and SPP-net are two typical deep learning-based methods leveraging multiscale strategies for UFZ classification. MSSN involves initially resampling HSR imagery to a uniform size of 512 × 512. Subsequently, deep features of images were extracted using a ResNet-101 network. To capture and integrate contextual semantic information, multiple atrous convolutions with varying dilation rates were applied to the deep feature map. By superimposing these contextual semantic features, the model obtained comprehensive and representative multiscale features within the feature maps, which were further employed for UFZ classification. For the SSP-Net, Liu et al. [43] successfully employed a multiscale network augmented with SPP layer for remote sensing scene classification, aiming to extract comprehensive multiscale features. Initially, they resampled the remote sensing images into three distinct sizes: 128 × 128, 192 × 192, and 256 × 256. Subsequently, three separate deep networks were utilized to extract deep feature maps corresponding to each resampled image size. The SPP layer, then, applied pooling operations at multiple scales to transform the deep feature maps of three sizes into feature vectors of equal length. Finally, leveraging multikernel learning techniques, the method aggregated the features extracted from different scales, thereby synthesizing a holistic representation of the remote sensing images.

In order to ensure a fair comparison, all five methods were tested on Zhuhai and Guangzhou datasets for UFZ classification, and evaluation results are shown in Tables IV and V. According to Table IV, in the Zhuhai study area, the proposed SSF-CNN achieved the highest OA, surpassing the second-best method (MSSN) by 2.65%, and a Kappa value exceeding the second best by 3.69%. Similarly, in the Guangzhou study area, the proposed method exhibited the highest OA (90.49%), outperforming the second-best method (MSSN) by 4.27% and a Kappa value exceeding the second best by 5.1%. Furthermore, the proposed method exhibited superior performance across various categories, as indicated by its higher $F1$ values.

The quantitative results were also corroborated in Figs. 12 and 13, which showed the comparison results of the five methods in typical areas of Zhuhai and Guangzhou, respectively. Visualizations in Figs. 12 and 13 showed that our proposed method achieved significantly better overall performance than the other four methods. It is noteworthy that the SO-CNN and DFN methods exhibit suboptimal performance in the green space and industrial categories, as indicated by

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TABLE V
QUANTITATIVE COMPARISONS IN THE GUANGZHOU DATASET

Method	F1 (%)							OA (%)	Kappa (%)
	Agricultural	Commercial	Industrial	Green space	Residential	Public service	Educational		
SO-CNN	81.08	74.57	69.70	62.07	75.17	69.42	82.35	75.66	69.99
DFN	84.85	90.91	77.97	59.26	83.52	83.33	90.32	81.81	77.58
MSSN	78.76	**94.94**	85.86	80.65	83.76	**93.98**	92.19	86.22	83.50
SPP-net	57.46	92.99	75.43	65.97	80.16	83.72	84.34	76.02	71.45
SSF-CNN	**86.91**	90.79	**91.15**	**85.71**	**94.12**	88.10	**94.12**	**90.49**	**88.60**

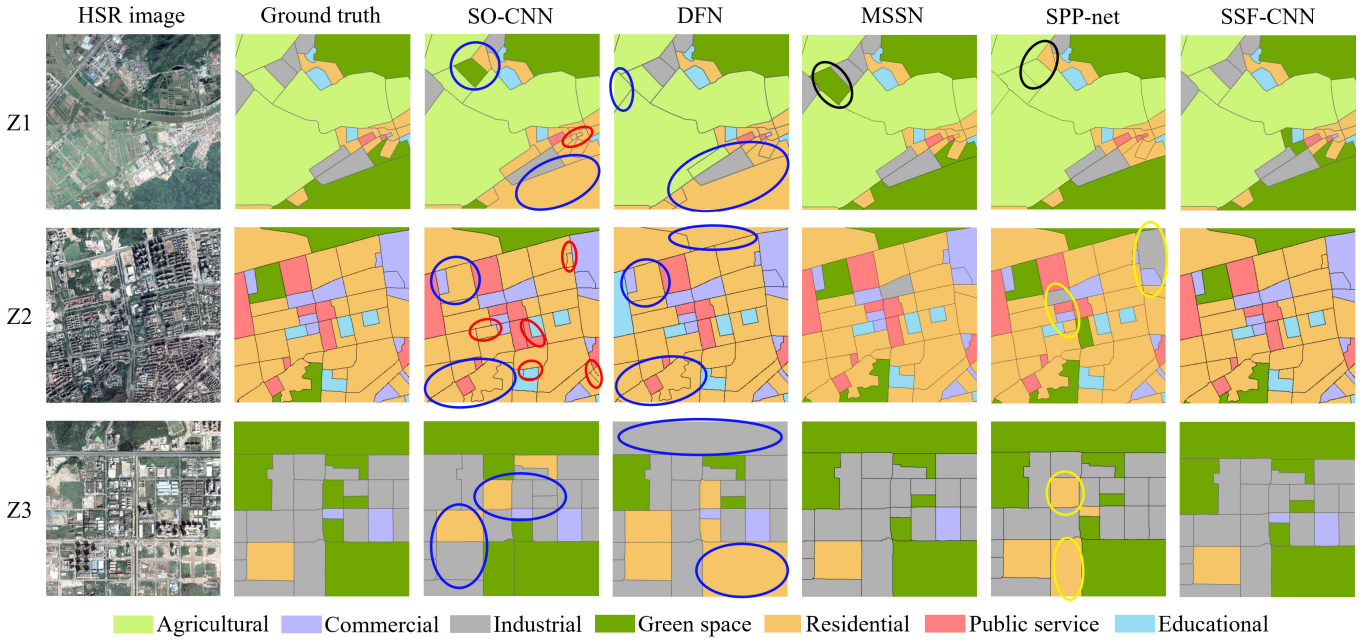


Fig. 12. Comparison results of different methods for UFZ classification in typical areas of Zhuhai.

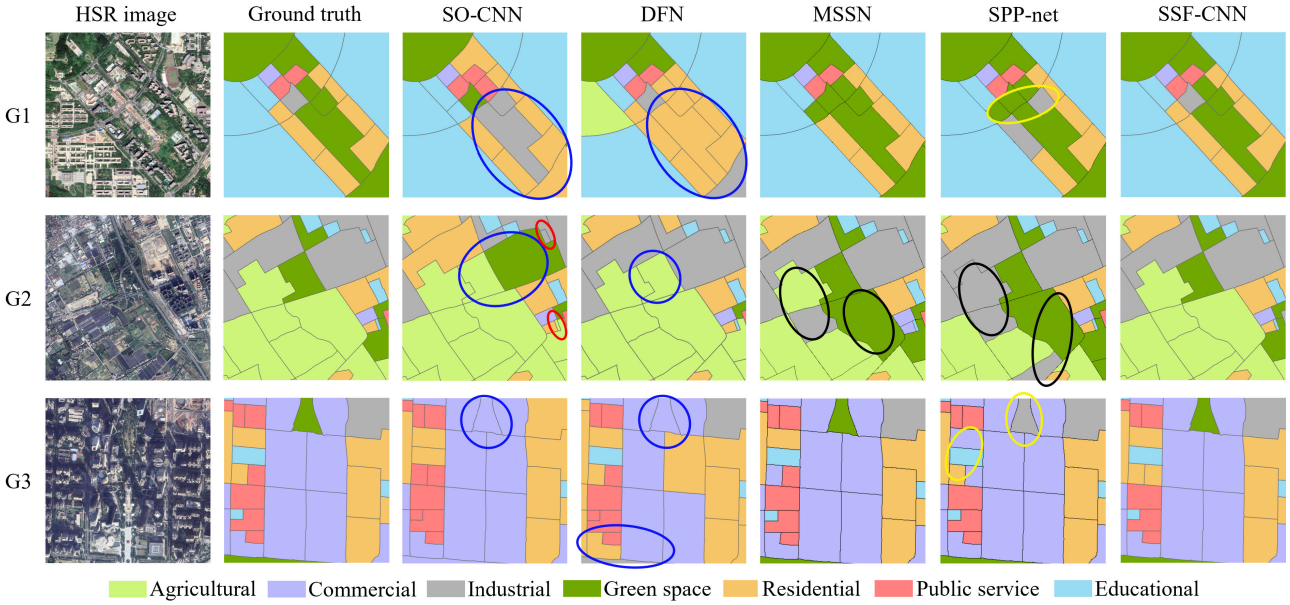


Fig. 13. Comparison results of different methods for UFZ classification in typical areas of Guangzhou.

the blue circles in Figs. 12 and 13. This can be attributed to the inability of the sampling methods to extract features across the entire UFZ effectively. An example of the green space category includes expansive regions within urban areas

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TABLE VI
QUANTITATIVE RESULTS OF UFZ CLASSIFICATION AFTER DISCARDING DIFFERENT POI CATEGORIES

Discarded POI categories	Commercial	Public service	Educational
OA (%)	84.38	85.42	86.46
Kappa (%)	80.41	81.68	83.05
Using all POIs (%)	OA = 87.85, Kappa = 84.68		

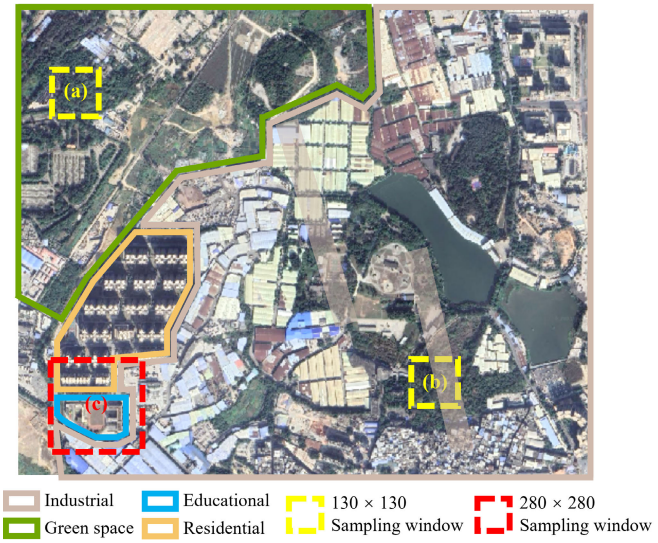


Fig. 14. Detailed display of sampling-based methods. (a) Sampling an industrial zone by the sampling window of 130 × 130. (b) Sampling a green space zone by the sampling window of 130 × 130. (c) Sampling an educational zone by the sampling window of 280 × 280.

covered with verdant vegetation, such as those found in ecological reserves. However, apart from the green space category, green areas can also be found in other UFZ categories, such as residential zones and industrial zones (an example can be found in Fig. 14). When using the sampling method presented in DFN, we may get many patches of green vegetation, which have totally different UFZ labels [see Fig. 14(a) and (b)]. Inputting such patches as training samples into CNN models will inevitably impact the model's ability to fit the green space category. Although the sampling methods provide a way to handle irregular and nonfixed-sized images, the shortcoming is also apparent, i.e., samples cannot represent the whole UFZ well.

One may argue that increasing the size of the sampling window can alleviate the issue. We have also tested it on the SO-CNN model. When the size of the sampling window is increased from 140 × 140 to 280 × 280, the classification accuracy of small UFZs noticeably decreases (red circles in Figs. 12 and 13). This is because the small UFZs occupy only a relatively small proportion in a large sampling window, which adds difficulties for the model to determine the category of a sampling patch. An example is shown in Fig. 14(c), which is an educational zone, and it is nearly impossible to use the above sampling approaches to identify its category.

Both MSSN and SPP-net exhibited subpar performance when applied to agricultural zones due to the considerable scale variations inherent in such regions (as indicated by the black circles in Figs. 12 and 13). Agricultural zones are characterized by small, varying-sized plots, which necessitates the extraction of multiscale features to identify them accurately. Therefore, a model with robust multiscale feature extraction capabilities becomes imperative. In this regard, the proposed SSF-CNN demonstrates notable classification accuracy, specifically in agricultural zones, thereby confirming the efficacy and superiority of its multiscale strategy.

In addition, for SPP-net, the sample sizes of the commercial, industrial, and educational zones of Zhuhai were smaller than those of Guangzhou. Consequently, the classification accuracy in these categories was also lower in Zhuhai. On the contrary, the sample sizes of the agricultural and green space zones of Guangzhou were smaller than those of Zhuhai, resulting in inferior classification accuracy for these categories in Guangzhou (as indicated by the yellow circles in Figs. 12 and 13). This observation underscores the sensitivity of SPP-net to sample size, as a decrease in sample size significantly impacts its classification accuracy.

VI. DISCUSSION

In this section, we mainly discussed the contributions of POI data to UFZ classification and the necessity of components in the proposed SSF-CNN model.

A. Contributions of POI Data Types

This study adopted three types of POI data to facilitate UFZ classification: commercial, public service, and educational POIs. In order to evaluate the contributions of each type of POI data, three ablation experiments were conducted in the Zhuhai dataset, among which one type of POI data was discarded, and the final UFZ classification results were evaluated. Table VI shows the quantitative results of UFZ classification after discarding POIs.

According to Table VI, both OA and Kappa were decreased after discarding any type of POIs, demonstrating that all POI data contributed positively to UFZ classification. The contributions of each POI category can be denoted by the decrease of OA after discarding it. Specifically, contributions of commercial, public service, and educational POIs were 3.47%, 2.43%, and 1.39%, respectively. Among the three POI types, commercial POIs had the highest contribution due to the similarity in texture structures between commercial and residential zones in HSR images, adding challenges to the distinguishment of the two UFZ types. By introducing commercial POIs, the capacity to distinguish commercial UFZs from residential UFZs was greatly enhanced. It is also noted that educational POIs have the lowest contribution, which can be explained by the fact that most educational UFZs have prominent interpretative marks, such as playgrounds and

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TABLE VII
RESULTS OF WITH/WITHOUT THE HGC MODULE

Conditions	OA (%)	Kappa (%)
Original SSF-CNN	87.85	84.68
Without HGC module	86.85	84.03

TABLE VIII
RESULTS OF PROCESSING THE ATTENTION MECHANISM UNDER DIFFERENT CONDITIONS

Conditions	OA (%)	Kappa (%)	Params
Original SSF-CNN	87.85	84.68	25.70M
Without the sCBAM module	86.83	83.69	25.70M
Substituting sCBAM with CBAM	87.85	84.75	28.22M

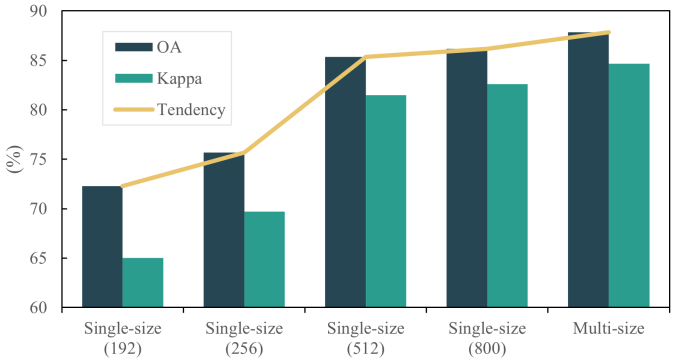


Fig. 15. Effect of image block size settings on classification accuracy.

sports fields, and HSR images alone can achieve comparatively high classification accuracies.

B. Necessities of the Components in SSF-CNN

SSF-CNN comprises four crucial components: the multiresolution resampling, the HGC module for multiscale feature learning, the attention mechanism in sCBAM, and the mask layer. To evaluate the necessity and efficacy of each component, experiments were conducted in the Zhuhai dataset by selectively removing or modifying each component and subsequently assessing influences on classification accuracies.

1) Efficacy of the Multiresolution Resampling: All images were resampled to multiple resolutions so that, when fed into the network for training, the image blocks were with multiple sizes, i.e., 192 × 192, 256 × 256, 512 × 512, and 800 × 800. Four additional sets of experiments were conducted to validate the effectiveness of this training method. In the first set of experiments, all blocks were resampled to a single size of 192 × 192. The same procedure was followed for the second to fourth sets of experiments, where the image blocks were resampled to 256 × 256, 512 × 512, and 800 × 800, respectively. During the prediction phase, the four single-size models adopted image blocks of the original size as inputs, which is consistent with the multisize model. The classification results of these four sets of experiments were compared with the original model (see Fig. 15).

As shown in Fig. 15, in terms of the single size, the classification accuracy is proportional to the size of the input image when the size is below 512. This may be attributed to the fact that larger image sizes can reduce the information loss when resampling all images to a uniform size. However, as the image size continues to increase (from 512 to 800), the increase in accuracy becomes less apparent, suggesting that simply increasing the image size may not be a practical solution for mitigating the loss of scale information. Upon the adoption of multisize images as input, a more substantial increase in accuracy was achieved. The multisize setting achieved higher classification accuracy than any single-size setting, thereby validating the effectiveness of multiresolution resampling in capturing scale information pertaining to UFZs.

2) Efficacy of the HGC Module: The HGC module can extract multiscale features from the input images of diverse sizes through hierarchical learning. An experiment was set up to verify the importance of this module. Specifically, the HGC module was removed, and the resulting classification accuracy was compared to that of the original model, as enumerated in Table VII.

As shown in Table VII, after removing the HGC module, the experiment achieved an OA of 86.85% and a Kappa of 84.68%, which were lower than those of the proposed SSF-CNN (87.85% in OA and 84.68% in Kappa). The decrease in OA and Kappa proved the efficacy of the HGC module.

3) Efficacy of the sCBAM Module: An sCBAM module with a spatial and channel attention mechanism was proposed and adopted to improve UFZ classification accuracies. At the same time, it reduces the number of parameters compared to the existing CBAM module. In order to evaluate the efficacy of the sCBAM module, two separate experiments were conducted. The first directly removed the sCBAM component, while the second substituted it with the traditional CBAM module. Classification accuracies and the number of parameters were compared with those of the original SSF-CNN, and the results were presented in Table VIII.

According to Table VIII, classification accuracies significantly dropped (1.02% in OA and 0.99% in Kappa) after removing the sCBAM module, which proved the efficacy of the sCBAM module in terms of improving classification accuracies. Results of the second experiment demonstrated that classification accuracies by CBAM were nearly the same as those by sCBAM. However, the number of parameters

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TABLE IX
RESULTS OF PROCESSING THE MASK LAYER UNDER DIFFERENT CONDITIONS

Model	OA (%)	Kappa (%)
Original SSF-CNN	87.85	84.68
Without mask layer	78.52	74.92
Mask layer before deep network	87.15	83.78

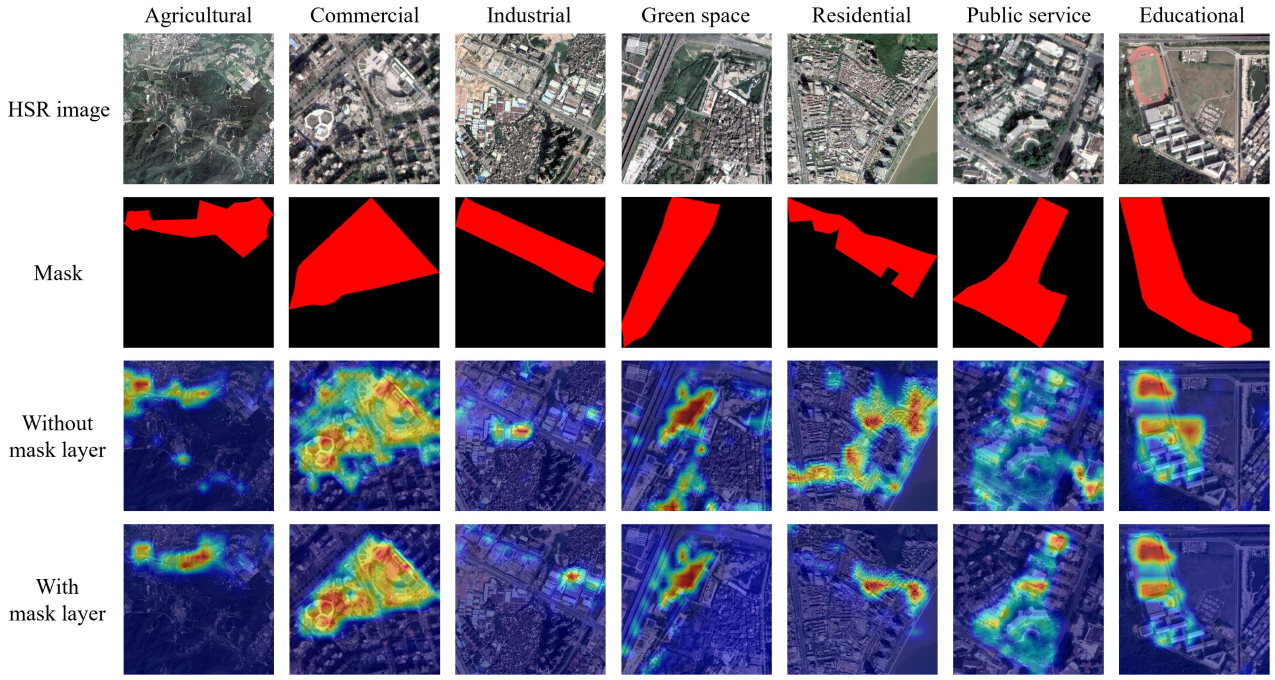


Fig. 16. Grad-CAM visualizations. (First row) Original HSR images of different UFZs. (Second row) Masks of UFZs. (Third row) CAM by the network without the mask layer. (Fourth row) CAM by the network with the mask layer.

of sCBAM is 9% less than that of the existing CBAM, which proved the efficacy of the sCBAM module in terms of decreasing parameter amounts.

4) Efficacy of the Mask Layer: The mask layer is adopted to fit the irregular shape of UFZs and is placed at the end of SSF-CNN after the deep feature learning network. In order to evaluate the efficacy of the mask layer, two additional experiments were conducted. The first one removed the mask layer of SSF-CNN and conducted UFZ classification without the mask layer. The second experiment placed the mask layer at the beginning of SSF-CNN before the deep feature learning network. The outcomes of the two experiments were subsequently compared with the original model, and the results were presented in Table IX.

It can be seen from Table IX that the model's classification accuracy significantly decreased after removing the mask layer, with OA and Kappa dropping by 9.33% and 9.76%, respectively. This is because the mask layer can help suppress irrelevant information outside the UFZ area. To intuitively and visually explain the efficacy of the mask layer, we used class activation maps (CAMs) to highlight the important regions for predicting UFZs. In this study, Grad-CAM [44] was employed to show outputs from several visualizations by networks with and without the mask layer (see Fig. 16). Red regions in Fig. 16 (rows 3 and 4) correspond to high scores where the proposed model highlights and blue regions correspond to regions which the proposed network paid less attention to. The results of Fig. 16 demonstrated that the mask layer could help models focus on the important regions, thus improving the ability to classify UFZs.

Furthermore, Table IX revealed that the position of the mask layer also matters. The OA and Kappa will decrease by 0.7% and 0.9% when the mask layer is shifted from the back of SSF-CNN to the beginning, respectively, before the deep network. This is because more contextual information can be learned from the surroundings of UFZs with the deep network when the mask layer is placed at the rear.

VII. CONCLUSION

This article proposes SSF-CNN, an end-to-end deep learning model that utilizes HSR images and POIs to capture physical and socioeconomic characteristics for improved classification of UFZs. It addresses two challenges posed by UFZ classification with deep learning methods: 1) sizes of UFZs vary greatly and 2) shapes of UFZs are irregular. To solve the above two challenges, the proposed model first adopts a novel network, named Res-HANet, which embeds an HGC structure and an attention mechanism to learn multiscale features from image blocks of four different sizes. Second, another novel network structure, i.e., a mask layer, is designed to filter the

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deep features so that only features within the irregular shape are retained and extraneous features are thus removed.

The proposed method was performed on two datasets: Zhuhai and Guangzhou datasets. The experimental results indicated that SSF-CNN demonstrated an OA of 87.85% and 90.49% on the Zhuhai and Guangzhou datasets, respectively. In addition, the Kappa of SSF-CNN attained 84.68% and 88.60% for the Zhuhai and Guangzhou datasets, respectively. We compared SSF-CNN with four existing CNN-based UFZ classification models: SO-CNN, DFN, MSSN, and SPP-net. From the results, SSF-CNN performed significantly better.

Moreover, the ablation experiments confirm the necessity of the multiple components of the proposed method. First, the selected three types of POIs can augment the classification outcomes. Second, removing the mask layer reduces classification accuracy, and superior classification results are attainable when the mask layer is placed after the deep network as compared to before it. Third, the classification accuracy will be influenced by the image size of the input network, whereby higher classification accuracy is achieved with larger input image sizes. However, resampling the image to multiple sizes increases classification accuracy, surpassing any single size. Fourth, HGC and sCBAM employed in the Res-HANet enhance the classification results. Moreover, the improved sCBAM involves fewer parameters than the original CBAM while attaining the same classification accuracy.

Furthermore, the model can be readily implemented in other study areas or datasets to facilitate large-scale UFZ mapping and extended to other classification scenarios.

Although the experimental results support the efficacy of the proposed SSF-CNN model, there are at least two limitations. First, the proposed model relies on a large number of precise training samples, which were mainly annotated manually. Second, this study simply employed road blocks of a single level as the basic unit for UFZ mapping, which may be inappropriate in some cases. In our future work, we aim to develop better algorithms to improve the performance of UFZ segmentation and explore methods of generating training samples automatically with auxiliary data.

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